

ECON2043 Experimental and Behavioural Economics

Seminar 5: Experimental Design

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Research Question

Discussion: Do you have any ideas? Please share.

- ▶ Interesting
- ▶ Clean design: one treatment and one control
- ▶ Clear outcome variables
- ▶ Doable in the classroom setting
- ▶ Related to AI

Please also think about time. We have 10 groups, and most likely we will show the QR codes one by one, so in practice we probably will not run two experiments at the same time.

Randomization

Suggested rule

- ▶ Before the experiment lecture, each participant will receive a unique experiment ID
- ▶ Use the last digit of the experiment ID
- ▶ Odd = control, even = treatment
- ▶ Or the other way round

Data Collection

In general, we recommend using Microsoft Forms to collect your data.¹

- ▶ You do not need to put every single step on the form
- ▶ The form can be used mainly to collect choices, ratings, and basic background variables
- ▶ Last year, some groups even designed their own software for participation

So our suggestion is: use a simple and robust tool to collect the data, unless you have a very clear reason not to.

¹Please ask for the experiment ID at the very beginning of every questionnaire. If you use the experiment ID for randomization, then the second question can ask for the last digit of the experiment ID, which can be used to separate treatment and control.

Example: Goal Setting

Research question

Does setting a goal improve subsequent performance?

Design

- ▶ Round 1: participants complete the task once
- ▶ Control group: move directly to Round 2
- ▶ Treatment group: set a goal before Round 2
- ▶ Round 2: participants complete the task again

Outcome variable

- ▶ Round 2 performance
- ▶ Or improvement from Round 1 to Round 2

Main idea

- ▶ Compare people who set a goal with people who do not

Possible Hypotheses

H1

Participants in the treatment group perform better in Round 2 than those in the control group.

H2

Within the treatment group, a higher goal is associated with better Round 2 performance.

Econometric Model (H1)

Pooled treatment effect

$$Y_i = \alpha + \beta \text{GoalSetting}_i + \gamma \text{Baseline}_i + \varepsilon_i$$

Notations

- ▶ Y_i : Round 2 performance of participant i
- ▶ GoalSetting_i : treatment dummy, equal to 1 if participant i is in the goal-setting group
- ▶ Baseline_i : Round 1 performance
- ▶ α : intercept
- ▶ β : pooled treatment effect
- ▶ γ : association between baseline and later performance
- ▶ ε_i : error term

Econometric Model (H2)

Within the treatment group

$$Y_i = \alpha + \beta \text{GoalLevel}_i + \gamma \text{Baseline}_i + \varepsilon_i$$

Notations

- ▶ Y_i : Round 2 performance of participant i
- ▶ GoalLevel_i : the goal chosen by participant i
- ▶ Baseline_i : Round 1 performance
- ▶ α : intercept
- ▶ β : association between the goal level and later performance
- ▶ γ : association between baseline and later performance
- ▶ ε_i : error term

General Econometric Model

Level form

$$Y_i = \alpha + \beta \text{Treat}_i + \gamma X_i + \varepsilon_i$$

Log form²

$$\log(Y_i) = \alpha + \beta \text{Treat}_i + \gamma X_i + \varepsilon_i$$

Interpretation

- ▶ In the level form, β is the difference in levels between treatment and control
- ▶ In the log form, β is approximately the percentage difference between treatment and control
- ▶ X_i can include baseline performance or other control variables

²Used when outcomes are very skewed or when percentage effects are easier to interpret than level effects.

Interpretation

If $\beta > 0$

- ▶ treatment is associated with a higher outcome

If $\beta < 0$

- ▶ treatment is associated with a lower outcome

In practice, use the simplest model that matches your outcome variable and your research question.

Demographics

Before you analyze your data, we will send a demographics questionnaire by email.

- ▶ It will include the experiment ID
- ▶ It may also include gender, age, AI attitude and other basic demographic information
- ▶ After we collect the responses, we will send the data to you

So you do not need to collect all demographic variables yourself during the classroom experiment.

PSet Q3: Choosing the Control

Question

- ▶ In a pre-test / treatment / post-test design, what should the control group do?

Main idea

- ▶ The control should depend on the claim you want to make
- ▶ If you want to compare your intervention with doing nothing, then “nothing” is a sensible control
- ▶ If you want to compare your intervention with another video, then that other video should be the control
- ▶ If you want to show that your intervention outperforms a competitor, then the competitor should be the control

Takeaway

- ▶ There is no universally correct control group
- ▶ The research question should guide the control choice

PSet Q4: Goal-Setting Regression Results

Question

- ▶ Estimate the treatment effect once without and once with controlling for the first attempt
- ▶ How does this affect the standard error of the treatment dummy?

Regression results

	Without control	With first attempt
Goal-setting dummy	-0.195 (0.268)	0.141 (0.083)
R^2	0.016	0.920

Takeaway

- ▶ The standard error falls sharply once we control for the first attempt
- ▶ This means the treatment effect is estimated more precisely
- ▶ The p-value for the goal-setting dummy changes from 0.473 to 0.101
- ▶ R^2 also rises strongly, because first attempt is highly predictive of second attempt

Appendix: Excel to CSV

Option 1: in Excel

- ▶ Open the Excel file
- ▶ Check that variable names are in the first row
- ▶ Click File → Save As
- ▶ Choose CSV UTF-8 (Comma delimited) (*.csv)
- ▶ Save the file, then import it into Stata

Option 2: in Stata

- ▶ `import excel "your_data.xlsx", firstrow clear`
- ▶ `export delimited using "your_data.csv", replace`

For most groups, Option 1 is simpler.

Appendix: Basic Stata Commands

Install package

```
ssc install estout, replace
```

Installs the package used to store and export regression results.

Import data and run OLS

```
import delimited "your_data.csv", clear
```

```
reg end_count goal_treat baseline_count, vce(robust)
```

```
reg end_count goal_gap baseline_count if goal_treat == 1, vce(robust)
```

- ▶ `import delimited`: imports a CSV file into Stata
- ▶ first `reg`: estimates the pooled treatment effect
- ▶ second `reg`: estimates the relationship within the treatment group
- ▶ `vce(robust)`: reports robust standard errors

Appendix: Exporting a Regression Table

Store models

```
eststo clear
eststo m1: reg end_count goal_treat baseline_count, vce(robust)
eststo m2: reg end_count goal_gap baseline_count ///
    if goal_treat == 1, vce(robust)
```

- ▶ `eststo clear`: clears previously stored models
- ▶ `eststo m1:` and `eststo m2::` store regression results

Display and export the table

```
esttab m1 m2, se star(* 0.10 ** 0.05 *** 0.01)
esttab m1 m2 using "results.rtf", replace ///
    se star(* 0.10 ** 0.05 *** 0.01)
```

- ▶ first `esttab`: shows the table in Stata
- ▶ second `esttab using`: exports the table to Word-readable `.rtf`
- ▶ `star(...)`: adds significance stars